

How to Assemble Your Own 3D Printer



ANKIT GUPTA

3D printing, getting popular for making usable objects, is the reverse of traditional machining where material is removed from a block by drilling, cutting, chiseling, etc for making objects. In 3D printing, an object is created by laying successive layers of material as per requirement. This article describes how you

can assemble a simple 3D printer for yourself, like we did in EFY Lab, and then make use of it. But before that, you should know some basics of 3D printing.

The process of printing 3D objects starts with making a virtual design of the object you want to create, using one of the supported computer aided design (CAD) software. A 3D scanner can also be used to copy an existing

object. The scanner makes a 3D digital copy of an object and puts it into a 3D modeling program. This 3D model file is sliced into thousands of horizontal layers which are then printed by a 3D printer, layer by layer, creating the entire 3D object. 3D printers generally employ one of below-mentioned methods.

Fused deposition modeling (FDM).

This is the most popular method used in DIY type 3D printers. Here a plastic filament or wire is made to pass through an extrusion nozzle (Fig. 2). The nozzle tip is heated to melt the filament or wire. The nozzle can be moved in all three directions, precisely, using stepper motors. The object is produced by extruding melted material to form layers upon layers of the object. The material hardens immediately after extrusion from the nozzle. The process is also known as fused filament fabrication (FFF). The printer shown in Fig. 1 is an assembled FDM based 3D printer.

Selective laser sintering (SLS). In this method high-power laser is used to fuse small particles of plastic, metal, ceramic or glass powder to form an object. The laser selectively scans and fuses the material layer by layer as per the 3D model design. Once a layer is

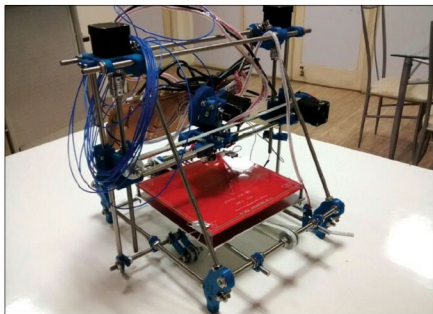


Fig. 1: LM8UU Prusa Mendel 3D printer

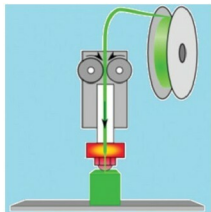


Fig. 2: Fused deposition modeling

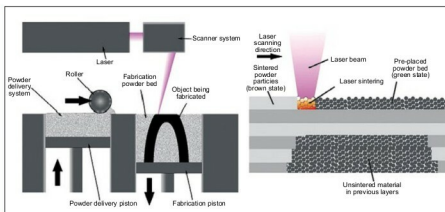


Fig. 3: SLS system